

LECTURE 1 (SEPTEMBER 2ND)

Definition. (Contour integral) Let C be a contour parametrised by $z = z(t)$ where $a \leq t \leq b$. Also, let f be continuous on C . Then, the contour integral of f is defined as

$$\int_C f(z) dz = \int_C u dx - v dy + i \int_C u dy + v dx$$

and also, through parametrisation, we have

$$\int_C f(z) dz = \int_a^b f(z(t)) z'(t) dt$$

Definition. (Derivative) The derivative of a complex function is defined as

$$f'(z_0) = \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0}$$

if it exists. If f is C^1 (the function is continuous up its first derivatives) and $u_x = v_y$ & $v_x = -u_y$, then $f'(z)$ exists.

Theorem. (Green's theorem) Let P and Q be a C^1 function on $\bar{D}$ where D is a simply connected bounded region whose boundary is C . Then,

$$\int_C P dx + Q dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy$$

where C is oriented in the positive sense. Here,

$$\int_C P dx = - \iint_D \frac{\partial Q}{\partial y} dy dx$$

and

$$\int_C Q dy = \iint_D \frac{\partial Q}{\partial x} dx dy$$

Proof. Notice that for a parametrised simple disc where the above is $y = g_2(x)$ and $y = g_1(x)$ on the bottom,

$$\int_C P dx = \int_a^b P(x, g_1(x)) dx - \int_a^b P(x, g_2(x)) dx$$

and

$$\iint_D \frac{\partial P}{\partial y} dy dx = \int_a^b \int_{g_1(x)}^{g_2(x)} \frac{\partial P}{\partial y} dy dx = \int_a^b \left[P(x, g_2(x)) - P(x, g_1(x)) \right] dx$$

Similarly,

$$\int_D Q \, dy = \int_c^d Q(y, h_2(x)) \, dy - \int_c^d Q(h_1(y), y) \, dy$$

and

$$\iint \frac{\partial Q}{\partial x} \, dx \, dy = \int_c^d \int_c^d \int_{h_1(y)}^{h_2(y)} \frac{\partial Q}{\partial x} \, dx \, dy = \left[Q(h_2(y), y) - Q(h_1(y), y) \right] dy$$

□

Theorem. (Cauchy's theorem) If f is analytic (differentiable) inside and on a bounded simply connected domain D whose boundary is C and $f \in C^1(\bar{D})$ then

$$\int_C f(z) \, dz = 0$$

Proof. Let $f = u + iv$. Then $u_x = v_y$ and $v_x = -u_y$ on D . Thus

$$\int_C f(z) \, dz = \int_C u \, dx - v \, dy + i \left[\int_C u \, dy + v \, dx \right]$$

where

$$\int_C u \, dx - v \, dy = \iint_D \left(-\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \, dx \, dy$$

and

$$\int_C u \, dy + v \, dx = \iint_D \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \, dx \, dy$$

notice how the terms in the large brackets vanish, and that the integral itself equals zero. □

Theorem. (Generalised Cauchy's Theorem) If $f \in C^1(\bar{D})$ is analytic in a bounded multiply connected domain D with finitely many holes inside, then

$$\int_{\partial D} f(z) \, dz = 0$$

where ∂D is a positively oriented boundary of D .

Theorem. (Cauchy Integral Formula) If f is analytic inside and on a POSCC C then for every z_0 inside C we have

$$f(z_0) = \frac{1}{2\pi i} \int_C \frac{f(z)}{z - z_0} \, dz$$

LECTURE 2 (SEPTEMBER 4TH)

Recall. (Cauchy's formula) For a positively oriented simple closed curve (POSCC) C and f that is analytic inside and on C , we have

$$f(z) = \frac{1}{2\pi i} \int_C \frac{f(\xi)}{\xi - z} d\xi$$

for all z inside C . If $\bar{D}(z_0, r) = \{z \mid |z - z_0| \leq r\} \subset D$,

$$\frac{1}{\xi - z} = \frac{1}{\xi - z_0 + z_0 - z} = \frac{1}{(\xi - z_0) \left(1 - \frac{z - z_0}{\xi - z_0}\right)}$$

for $z \in D(z_0, r)$. However,

$$\left| \frac{z - z_0}{\xi - z_0} \right| < \frac{|z - z_0|}{r} < 1$$

so that

$$\frac{1}{\xi - z} = \frac{1}{\xi - z_0} \sum_{n=0}^{\infty} \left(\frac{z - z_0}{\xi - z_0} \right)^n = \sum_{n=0}^{\infty} \frac{(z - z_0)^n}{(\xi - z_0)^{n+1}}$$

Thus

$$\begin{aligned} f(z) &= \frac{1}{2\pi i} \int_C f(\xi) \sum_{n=0}^{\infty} \frac{(z - z_0)^n}{(\xi - z_0)^{n+1}} d\xi \\ &= \sum_{n=0}^{\infty} a_n (z - z_0)^n \end{aligned}$$

where

$$a_n = \frac{1}{2\pi i} \int_C \frac{f(\xi)}{(\xi - z_0)^{n+1}} d\xi = \frac{1}{n!} f^{(n)}(z_0)$$

Notice that the convergence is uniform for $\xi \in C$.

Theorem. If f_n is continuous in C such that $f_n \rightarrow f$ uniformly on C

$$\int_C f(z) dz = \lim_{n \rightarrow \infty} \int_C f_n(z) dz$$

Theorem. If $f(z) = \sum_{n=0}^{\infty} c_n (z - z_0)^n$ for $z \in D(z_0, R)$ then for $0 < r < R$ we have

$$\frac{1}{2\pi} \int_0^{2\pi} |f(z_0 + re^{it})|^2 dt = \sum_{n=0}^{\infty} |c_n|^2 r^{2n}$$

Proof. Take

$$f(z_0 + re^{it}) = \sum_{n=0}^{\infty} c_n r^n e^{int}$$

so that

$$\begin{aligned}
\frac{1}{2\pi} \int_0^{2\pi} |f(z_0 + re^{it})|^2 dt &= \frac{1}{2\pi} \int_0^{2\pi} \sum_{n=0}^{\infty} c_n r^n e^{int} \sum_{m=0}^{\infty} \bar{c}_m r^m e^{-imt} dt \\
&= \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} \frac{1}{2\pi} \int_0^{2\pi} c_n \bar{c}_m r^{n+m} e^{i(n-m)t} dt \\
&= \sum_{n=0}^{\infty} |c_n|^2 r^{2n}
\end{aligned}$$

Here, we note that

$$\int_0^{2\pi} e^{i(n-m)t} dt = \begin{cases} 2\pi & n = m \\ 0 & n \neq m \end{cases}$$

□

Corollary. (Liouville theorem) All bounded entire functions are constant. Using this theorem, we can prove the fundamental theorem of algebra.

Proof. If f is entire such that $|f(z)| \leq M$ for all $z \in \mathbf{C}$ then

$$\sum_{n=0}^{\infty} |c_n|^2 r^{2n} \leq M^2$$

for all $r > 0$ which means that $c_1 = c_2 = \dots = 0$.

□

Theorem. (Identity theorem) Let D be a domain (open and connected) and $\{z_n\}$ be a sequence of distinct points in D such that $z_n \rightarrow z_0 \in D$. If f is analytic in D with $f(z_n) = 0$ for all $n \in \mathbf{N}$, then $f(z) = 0$ for all $z \in D$.

Theorem. (Maximum modulus theorem) Let f be analytic in a domain D . Then, $|f|$ can not have a local maximum in D unless it is constant.

Proof. If

$$|f(z_0 + re^{it})| \leq |f(z_0)|$$

for all $t \in [0, 2\pi]$, then

$$\sum_{n=0}^{\infty} |c_n|^2 r^{2n} \leq |f(0)|^2 = |c_0|^2$$

which means that $c_1 = \dots = c_n = 0$. Then we can apply the identity theorem.

□

Theorem. (Morera's theorem) Morera's theorem can be thought as Cauchy theorem's inverse. If f is continuous in a domain D such that

$$\int_C f(z) dz = 0$$

for all closed contours C in D , then f is analytic in D (note that this also means that the integral is path independent).

Proof. The key point is that for a given $a \in D$, $F(z) = \int_a^z f(\xi) d\xi$ is well defined. Take any $z_0 \in D$. We'll show that $F'(z_0) = f(z_0)$, which means that $f = F'$ is analytic. Take any $\varepsilon > 0$. We choose $\delta > 0$ so that

(i) $D(z_0, \delta) \subset D$

(ii) If $|z - z_0| < \delta$ then $|f(z) - f(z_0)| < \varepsilon$

We need to prove that if $0 < |z - z_0| < \delta$ then

$$\left| \frac{F(z) - F(z_0)}{z - z_0} - f(z_0) \right| < \varepsilon$$

Take the line segment path $[z_0, z]$ from z_0 to z . Then

$$F(z) - F(z_0) = \int_{[z_0, z]} f(\xi) d\xi$$

and

$$\frac{F(z) - F(z_0)}{z - z_0} - f(z_0) = \frac{1}{z - z_0} \int_{[z_0, z]} [f(\xi) - f(z_0)] d\xi$$

Notice that now, due to (ii), the term inside the square brackets can be reduced to less than ε since $|\xi - z_0| < \delta$. Therefore

$$\left| \frac{F(z) - F(z_0)}{z - z_0} - f(z_0) \right| < \frac{1}{|z - z_0|} |z - z_0| \varepsilon = \varepsilon$$

□

Theorem. Let $\{f_n\}$ be a sequence of analytic functions in a domain D such that $f_n \rightarrow f$ uniformly on every compact subset of D . Then f is analytic on D .

Example. On $\mathbf{R}$, consider

$$f_n(x) = \sum_{k=1}^n \frac{\cos 3^k x}{2^k} \rightarrow f(x) = \sum_{n=1}^{\infty} \frac{\cos 3^n x}{2^n}$$

where the convergence is uniform and the right-hand side is nowhere differentiable.

Proof. First, f is continuous on D since convergence is uniform. Let C be any closed contour in D . Then

$$\int_C f(z) dz = \lim_{n \rightarrow \infty} \int_C f_n(z) dz = 0$$

By Morera's theorem, the proof is complete. $\square$

LECTURE 3 (SEPTEMBER 9TH)

Lemma. Let f be analytic on a simply connected domain D such that $f(z) \neq 0$ for any $z \in D$. Then,

- (i) There is g analytic on D such that $e^g = f$ on D
- (ii) There is h analytic on D such that $h^2 = f$ on D (this implies the Riemann mapping theorem)

Proof. (1) By Cauchy's theorem, the $\int_C f(z) dz = 0$ for every closed contour C in D . Thus for a given $z_0 \in D$, the integral

$$\int_{z_0}^z f(\xi) d\xi$$

is independent of path. Since f has no zero in D , f'/f is analytic on D . We define g on D by

$$g(z) = \int_{z_0}^z \frac{f'(\xi)}{f(\xi)} d\xi + \text{Log } f(z_0)$$

Then we'll show that $f(z) = e^{g(z)}$ on D . Observe that

$$\begin{aligned} (fe^{-g})' &= f'e^{-g} + f(e^{-g})' \\ &= f'e^{-g} + fe^{-g}\left(-\frac{f'}{f}\right) \\ &= f'e^{-g} - f'e^{-g} = 0 \end{aligned}$$

on D and $fe^{-g} = c$ a constant. If we define $h(z) = f(z)e^{-g(z)}$ then evaluating $h(z)$ at z_0 gives

$$\begin{aligned} h(z_0) &= f(z_0)e^{-g(z_0)} \\ &= f(z_0)e^{-\text{Log } f(z_0)} \\ &= f(z_0)[f(z_0)]^{-1} = 1 \end{aligned}$$

Therefore, $f(z) = e^{g(z)}$.

(2) By (1), there exists an analytic g such that $f = e^g$ on D . If we define $h = e^{g/2}$, then h is analytic on D with $h^2 = f$. $\square$

Theorem. (Exhaustion of D by compact subsets) Let $D \subset \mathbf{C}$ be open. Then there is a sequence of compact subsets K_n of D such that

(i)

$$K_1 \subset K_2 \subset \dots \subset K_n \subset D$$

(ii)

$$\bigcup_{n=1}^{\infty} K_n = D$$

(iii) If $K \subset D$ is compact then $K \subset K_n$ for some $n \in \mathbf{N}$

(iv) There is a sequence $\{\delta_n\}$ with $\delta_n > 0$ such that

$$\bar{D}(z, 2\delta_n) \subset K_{n+1}$$

for each $z \in K_n$

Proof. Define

$$K_n = \left\{ z \in \mathbf{C} \mid |z| \leq n \right\} \cap \left\{ z \in \mathbf{C} \mid d(z, D^c) \geq \frac{1}{n} \right\}$$

$$G_n = \left\{ |z| < n \right\} \cap \left\{ d(z, D^c) > \frac{1}{n} \right\}$$

and $G_n = D$ by the Archimedean property. □

LECTURE 4 (SEPTEMBER 16TH)

Theorem. Let D be a domain in $\mathbf{C}$. There is a sequence $\{K_n\}$ of compact subsets of D and $\{\delta_n\}$ ($\delta_n > 0$) such that

(i)

$$\bigcup_n K_n = D$$

(ii) If $K \subset D$ is compact then $K \subset K_n$ for some n

(iii) $\bar{D}(z, 2\delta_n) \subset K_{n+1}$ for every $z \in K_n$, $n = 1, 2, \dots$

Note that $z \notin F$ and F is closed. This implies that $d(z, F) = \min\{|z - w| \mid w \in F\} > 0$.

Proof. Put

$$K_n = \left\{ |z| \leq n \right\} \cap \left\{ d(z, D^c) \geq \frac{1}{n} \right\}$$

□

Example. Consider

$$f_n(x) = \frac{\sin(nx)}{\sqrt{n}} \rightarrow 0$$

which converges uniformly on $\mathbf{R}$. However,

$$f'_n(0) = \sqrt{n} \rightarrow \infty$$

as $n \rightarrow \infty$.

Theorem. If $\{f_n\}$ is a sequence of analytic functions on a domain D such that $f_n \rightarrow f$ uniformly on every compact subset of D , then

- (i) f is analytic on D (by Morera's theorem)
- (ii) $f'_n \rightarrow f'$ uniformly on every compact subset of D

Proof. Take any compact subset K of D . Then $K \subset K_n$ for some n , and $\bar{D}(z, \delta_n) \subset K_{n+1}$ for all $z \in K$. So, there exists $r > 0$ and K' , a compact subset of D , such that $\bar{D}(z, r) \subset K'$ for all $z \in K$. Here, applying Cauchy's integral formula,

$$f(z) = \frac{1}{2\pi i} \int_{|\xi-z|=r} \frac{f(\xi)}{(\xi-z)^2} d\xi$$

For every $z \in K$, we have

$$f'(z) = \frac{1}{2\pi i} \int_{|\xi-z|=r} \frac{f(\xi)}{(\xi-z)^2} d\xi$$

which implies that

$$|f'(z)| \leq \frac{1}{2\pi} \frac{\|f\|_{K'}}{r^2} (2\pi r) = \frac{\|f\|_{K'}}{r}$$

with $\|f\|_K = \max_{z \in K} |f(z)|$. Therefore, for every $z \in K$,

$$|f'_n(z) - f'(z)| \leq \frac{\|f_n - f\|_{K'}}{r}$$

which means $f'_n \rightarrow f'$ uniformly on K . □

Example. (Linear mapping) Consider the linear mapping

$$w = az + b$$

This mapping we see that, by expressing $a = re^{i\theta}$, the map stretches, rotates, and translates the complex plane.

Definition. (Linear functional transformation) For $ad - bc \neq 0$,

$$w = T(z) = \frac{az + b}{cz + d}$$

This can be written as

$$w = T(z) = \frac{a}{c} + \frac{bc - ad}{c} \cdot \frac{1}{cz + d}$$

From this, we understand that the map is 1-1 as the following inverse map exists:

$$z = \frac{dw - b}{-cw + a}$$

Additionally, we see that the mapping can be seen as a combination of translation, dilation ($|a| > 1$) or contraction ($|a| \leq 1$), rotation, and inversion.

$$z \rightarrow \frac{1}{z}$$

This is the tricky one!

Remark. (Inversion map $w = 1/z$) The equation (locus) of all lines and all circles in $\mathbf{R}^2$ is

$$a(x^2 + y^2) + bx + cy + d = 0$$

where $b^2 + c^2 > 4ad$ and $a, b, c, d \in \mathbf{R}$. In the complex plane, for $x^2 + y^2 = |z|^2$, $x = (z + \bar{z})/2$ and $y = (z - \bar{z})/2i$ so that

$$a|z|^2 + \frac{1}{2}(b - ic)z + \frac{1}{2}(b + ic)\bar{z} + d = 0$$

and again $b^2 + c^2 > 4ad$. This implies that

$$a|z|^2 + \beta z + \bar{\beta}\bar{z} + d = 0$$

where $a, d \in \mathbf{R}$ and $\beta \in \mathbf{C}$ with $|\beta|^2 > ad$. Taking the inversion map,

$$a \frac{1}{|w|^2} + \beta \frac{1}{w} + \bar{\beta} \frac{1}{\bar{w}} + d|w|^2 = 0$$

and

$$a + \beta\bar{w} + \bar{\beta}w + d = 0$$

by multiplying $|w|^2 = w\bar{w}$ ($\beta\bar{\beta} > ad$). We therefore see that the inversion map takes the set of all lines and circles to again the set of all lines and circles.

Definition. (S^2) S^2 is defined as the boundary of a unit ball in $\mathbf{R}^3$. We see this as a one-point compactification (Alexandroff compactification) of $\mathbf{C}$, and is seen as $\mathbf{C} \cup \{\infty\}$ where $\{\infty\}$ is called the north pole.

Lemma. The linear functional transformation which takes $\{a, b, c\}$ to $\{0, 1, \infty\}$ is uniquely determined by

$$T(z) = \frac{(z-a)(b-c)}{(z-c)(b-a)}$$

LECTURE 5 (SEPTEMBER 18TH)

Definition. Notice that the linear fractional transformation defined for $ad - bc \neq 0$ as

$$T(z) = \frac{az + b}{cz + d}$$

we see that the derivative is

$$T'(z) = \frac{ad - bc}{(cz + d)^2} \neq 0$$

and that it is analytic and T' is never 0. Meanwhile,

$$T^{-1}(z) = \frac{dz - b}{-cz + a}$$

Conformal are by definition, angle preserving maps (both size and orientation).

Definition. Let $z = z(t)$ for $a \leq t \leq b$ represent the contour C . Also, let $w = f(z)$, that is, $w(t) = f(z(t))$ represent the contour Γ . If $z_0 = z(t_0)$ and $w_0 = f(z_0) = f(z(t_0))$, f is analytic at z_0 such that $f'(z_0) \neq 0$. Then, from

$$w'(t_0) = f'(z_0)z'(t_0)$$

we have

$$\arg w'(t_0) = \arg f'(z_0) + \arg z'(t_0)$$

Thus, we see that analytic functions with nonzero derivatives are conformal mappings.

Definition. Consider

$$T(z) = \frac{az + b}{cz + d}$$

(with $ad - bc \neq 0$) and

$$S(z) = \frac{ez + f}{gz + h}$$

(with $eh - gf \neq 0$) we ultimately have

$$(T \circ S)(z) = \frac{(ae + bg)z + (af + bh)}{(ce + dg)z + (cf + dh)}$$

also,

$$T^{-1}(z) = \frac{dz - b}{-cz + a}$$

Ultimately, we see that these mappings are equivalent to matrix multiplication.

Proposition. (The fixed points of $T(z) = z$) We see that

$$\frac{az + b}{cz + d} = z \quad \implies \quad cz^2 + (d - a)z - b = 0$$

and that there can't be more than two fixed points. If S and T are linear fractional transformations such that $S(z_k) = T(z_k)$ for $k = 1, 2, 3$, then $S = T$ since $S^{-1} \circ T = I$.

Theorem. If distinct points $\{z_1, z_2, z_3\}$ and distinct points $\{w_1, w_2, w_3\}$ are points of $S^2 = \mathbf{C} \cup \{\infty\}$ then there exists a unique linear fractional transformation T such that $T(z_k) = w_k$ for $k = 1, 2, 3$.

$$\{a, b, c\} \rightarrow \{0, 1, \infty\} \leftrightarrow \{\alpha, \beta, \gamma\}$$

Then

$$T(z) = \frac{(b - c)(z - a)}{(b - a)(z - c)} \quad \& \quad S(z) = \frac{(\beta - \gamma)(z - \alpha)}{(\beta - \alpha)(z - \gamma)}$$

Example. Find V such that

$$\{0, 1, 2\} \rightarrow \{-1, 0, 4\}$$

Proof.

$$S(z) = \frac{-4(z + 1)}{1(z - 4)} = \frac{-4z - 4}{z - 4}$$

(we say it maps to $\{0, 1, \infty\}$) which is equivalent to

$$\begin{pmatrix} -4 & -4 \\ 1 & -4 \end{pmatrix}$$

For T that sends $\{0, 1, 2\}$ to the same points, we find the matrix, and $V(z)$ is given as

$$V(z) = \frac{8z - 8}{-3z + 8}$$

□

Example. Let

$$T(z) = \frac{1 + z}{1 - z}$$

which corresponds to

$$\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$$

Let $U = \{|z| < 1\}$, and say that we want to find $T(U)$. Notice that

$$\{-1, 0, 1\} \mapsto \{0, 1, \infty\}$$

As $T([-1, 1])$ maps to the positive x -axis, we can see that $T(-1) = \infty$ and that $T(|z| = 1)$ is a line passing $T(-1) = 0$. Then, $T(i) = i$ and $T(|z| = 1)$ is the y -axis. Another way to do this is through conformality of $T(|z| = 1)$ and the positive real axis being $\pi/2$ at $z = 0$.

Definition. Let U be the unit disk and $T = \{|z| = 1\}$ be a unit circle. For $\alpha \in U$, we define

$$\phi_\alpha(z) = \frac{z - \alpha}{1 - \bar{\alpha}z} : \begin{pmatrix} 1 & -\alpha \\ -\bar{\alpha} & 1 \end{pmatrix}$$

Then ϕ_α is analytic on $\bar{U}$ such that $\phi_\alpha^{-1} = \phi_{-\alpha}$, $\phi_\alpha(0) = -\alpha$, $\phi_\alpha(\alpha) = 0$, $\phi_\alpha(U) = U$, $\phi_\alpha(T) = T$, and $\phi'_\alpha(0) = 1 - |\alpha|^2$, $\phi'_\alpha(\alpha) = 1/(1 - |\alpha|^2)$.

Proof. We'll show that $\phi_\alpha(T) = T$.

$$\begin{aligned} \phi_\alpha(e^{it}) &= \frac{e^{it} - \alpha}{1 - e^{-it}\bar{\alpha}} = \frac{e^{it}(1 - e^{-it}\alpha)}{1 - e^{-it}\bar{\alpha}} \\ |\phi_\alpha(e^{it})| &= 1 \end{aligned}$$

□

LECTURE 6 (SEPTEMBER 23RD)

Recall. Last class, we've took $U = \{|z| < 1\}$ and $T = \{|z| = 1\}$ with $\alpha \in U$. The map $\phi_\alpha(z)$ was defined as

$$\phi_\alpha(z) = \frac{z - \alpha}{1 - \bar{\alpha}z} : \begin{pmatrix} 1 & -\alpha \\ -\bar{\alpha} & 1 \end{pmatrix}$$

We've noted that $1 - \bar{\alpha}z = 0$ implied that $z = \overline{(1/\alpha)} \notin \bar{U}$. You see that ϕ_α is analytic and one-to-one. We find explicitly that

$$\phi_\alpha^{-1} = \begin{pmatrix} 1 & \alpha \\ \bar{\alpha} & 1 \end{pmatrix}$$

such that $\phi_\alpha^{-1} = \phi_{-\alpha}$

$$\phi_\alpha^{-1}(z) = \phi_{-\alpha} = \frac{z + \alpha}{1 + \bar{\alpha}z}$$

and

$$\phi'_\alpha(z) = \frac{1 - |\alpha|^2}{(1 - \bar{\alpha}z)^2}$$

so that $\phi'_\alpha(0) = 1 - |\alpha|^2$ and $\phi'_\alpha(\alpha) = 1/(1 - |\alpha|^2)$. What about mappings from T ? If $z = e^{it}$, then

$$\phi_\alpha(e^{it}) = \frac{e^{it} - \alpha}{1 - \bar{\alpha}e^{it}} = \frac{e^{it} - \alpha}{e^{it}(e^{-it} - \bar{\alpha})} = \frac{1}{e^{it}} \frac{e^{it} - \alpha}{(e^{it} - \alpha)}$$

such that $|\phi_\alpha(e^{it})| = 1$ and $\phi_\alpha(T) \subset T$. Furthermore, we know that $\phi_{-\alpha}(T) \subset T$. We thus find, that $\phi_\alpha(T) = T$. With this information, via the maximum modulus theorem, we find that

$$\max_{z \in \bar{U}} |\phi_\alpha(z)| = \max_{z \in T} |\phi_\alpha(z)| = 1$$

which implies that

$$|\phi_\alpha(z)| < 1$$

for all $z \in U$.

Remark. We summarize like the following:

(i) For $\alpha \in U$,

$$\phi_\alpha(z) = \frac{z - \alpha}{1 - \bar{\alpha}z}$$

(ii) $\phi_\alpha : U \rightarrow U$ is one-to-one, analytic and onto.

(iii) $\phi_\alpha(T) = T$

(iv) $\phi'_\alpha(0) = 1 - |\alpha|^2$

(v) $\phi'_\alpha(\alpha) = 1/(1 - |\alpha|^2)$

Theorem. (Schwartz lemma) Let $f : U \rightarrow U$ is analytic with $f(0) = 0$. For every $z \in U$, we have $|f(z)| \leq |z|$ and $|f'(0)| \leq 1$. Furthermore, if one of the equalities hold, for some $z \in U \setminus \{0\}$ then f is a rotation.

$$f(z) = e^{i\theta} z$$

for some θ .

Proof. Notice that there exists a g analytic on U such that $f(z) = zg(z)$. For every fixed $z \in U$,

$$|g(z)| = \left| \frac{f(z)}{z} \right| \leq \max_{\theta} \left| \frac{f(re^{i\theta})}{r} \right| \leq \frac{1}{r}$$

if $|z| < r < 1$. Taking $r \rightarrow 1^-$, $|f(z)/z| \leq 1$. That is, $|f(z)| \leq |z|$ for all $z \in U$. Also, $f'(z) = g(z) + zg'(z)$ which implies that $f'(0) = g(0)$. This again implies that $|f'(0)| = |g(0)| \leq 1$.

Notice the following. If $f(0) = \alpha$, then define $g(z) = \phi_\alpha(f(z))$. Then, $g : U \rightarrow U$ is analytic

with $g(0) = 0$. Applying the Schwartz lemma to g , $|g'(0)| \leq 1$ implies that

$$|f'(0)| \leq 1 - |\alpha|^2$$

But,

$$g'(0) = \phi'_\alpha(f(0))f'(0) = \phi'_\alpha(\alpha)f'(0) = \frac{1}{1 - |\alpha|^2}f'(0)$$

□

Theorem. (Pick theorem) If $f : U \rightarrow U$ is analytic with $f(\alpha) = \beta$, then

$$|f'(\alpha)| \leq \frac{1 - |\beta|^2}{1 - |\alpha|^2}$$

The equality holds when

$$f(z) = \phi_{-\beta}(e^{i\theta}\phi_\alpha(z))$$

Proof. We define g on U so that $g(0) = 0$. Indeed,

$$g = \phi_\beta \circ f \circ \phi_{-\alpha}$$

is analytic from U into U with $g(0) = 0$. Applying the Schwartz lemma to g , we have $|g'(0)| \leq 1$. But

$$g'(0) = \left(\phi'_\beta[f(\phi_\alpha(0))] \right) f'(\phi_\alpha(0)) \phi'_\alpha(0) = \phi'_\beta(\beta) \cdot f'(0) \cdot \phi'_{-\alpha}(0) = \frac{1 - |\alpha|^2}{1 - |\beta|^2} f'(0) \leq 1$$

When does the equality hold? Let $g(z) = \lambda z$ for some $|\lambda| = 1$. Then

$$\phi_\beta(f(\phi_\alpha(z))) = \lambda z$$

By putting $z \rightarrow \phi_\alpha(z)$,

$$\phi_\beta(f(z)) = \lambda \phi_\alpha(z)$$

and

$$\phi_{-\beta}[\phi_\beta(f(z))] = \phi_{-\beta}(\lambda \phi_\alpha(z))$$

and ultimately

$$f(z) = \phi_{-\beta}(\lambda \phi_\alpha(z))$$

□

Theorem. If f is one-to-one analytic from U onto U , then there exists $\alpha \in U$ and $\theta \in \mathbf{R}$ such that $f(z) = e^{i\theta}\phi_\alpha(z)$ for all $z \in U$.

LECTURE 7 (SEPTEMBER 25TH)

Theorem. (Schwartz-Pick theorem) Let $f : U \rightarrow U$ be an analytic function such that $f(\alpha) = \beta$. Then,

$$|f'(\alpha)| \leq \frac{1 - |\beta|^2}{1 - |\alpha|^2}$$

and the equality holds when

$$f(z) = \phi_{-\beta}(\lambda \phi_{\alpha}(z))$$

for some $\lambda \in T$.

Example. Observe that $f(x) = x^3$ is one-to-one on $\mathbf{R}$, but $f'(0) = 0$. Therefore, we see that on $\mathbf{R}$, f being one-to-one on an interval I doesn't necessarily imply that $f'(x) \neq 0$ for any $x \in I$. However, the converse works as if $f'(x) \neq 0$ for any $x \in I$, then f is one-to-one by the mean value theorem.

However, on $\mathbf{C}$, $f(z) = e^z$ satisfies $f'(z) \neq 0$ for any $z \in \mathbf{C}$ but f is not one-to-one. We claim that if f is analytic and one-to-one on a domain D , then $f'(z) \neq 0$ for any $z \in D$.

Theorem. If $f : U \rightarrow U$ is one-to-one analytic, meaning that $f'(z) \neq 0$ for any $z \in U$, then $f(z) = e^{i\theta} \phi_{\alpha}(z)$ for some $\alpha \in U$, $\theta \in \mathbf{R}$

Proof. Let $f(\alpha) = 0$. First define $g = f^{-1}$ on U . Then $g(f(z)) = z$ for all $z \in U$ and g is analytic on U . Due to the fact that the derivative is never zero, we can use the inverse function theorem, where we are able to write

$$\frac{g(w) - g(w_0)}{w - w_0} = \frac{z - z_0}{f(z) - f(z_0)} = \frac{1}{\frac{f(z) - f(z_0)}{z - z_0}}$$

and

$$g'(w_0) = \frac{1}{f'(z_0)}$$

since $f'(z_0) \neq 0$. We see that

$$g'(0)f'(\alpha) = 1$$

by the chain rule. By the Schwartz-Pick theorem,

$$|f'(\alpha)| \leq \frac{1}{1 - |\alpha|^2}$$

and $|g'(0)| \leq 1 - |\alpha|^2$. This means that

$$|f'(\alpha)| = \frac{1}{1 - |\alpha|^2}$$

By the Schwartz lemma,

$$f(z) = \lambda \phi_\alpha(z)$$

for some $\lambda \in T$. □

Theorem. If f is analytic in a domain D with $f'(z_0) \neq 0$ for some $z_0 \in D$ then there is $r > 0$ such that $D(z_0, r) \subset D$ and f is one-to-one on $D(z_0, r)$.

Proof. If $z, w \in D(z_0, r)$ and $z \neq w$, then

$$\left| \frac{f(z) - f(w)}{z - w} - f'(z_0) \right| < \frac{1}{2} |f'(z_0)|$$

which implies that

$$\frac{1}{2} |f'(z_0)| < \left| \frac{f(z) - f(w)}{z - w} \right|$$

if $z, w \in D(z_0, r)$ with $z \neq w$. □

Lemma. If f is analytic in a domain D and

$$g(z, w) = \begin{cases} \frac{f(z) - f(w)}{z - w} & z \neq w \\ f'(z) & z = w \end{cases}$$

then g is continuous on $D \times D$. Note that this is trivially true on $\mathbf{R}$.

Lemma. If $|b - a| < \frac{|a|}{2}$, then $|b| > \frac{|a|}{2}$.

Proof. Notice that it suffices to prove the statement only when $z = w$. Since f' is continuous on D , for every ε , there exists a δ such that if $z \in D(a, \delta)$ then $|f'(z) - f'(a)| < \varepsilon$. Take any z, w ($z \neq w$) in $D(a, r)$. Define $\xi(t) = tz + (1 - t)w$ for $0 \leq t \leq 1$. Then,

$$\int_0^1 f'(\xi(t)) dt = \frac{f(z) - f(w)}{z - w} = g(z, w)$$

so that

$$g(z, w) - g(a, a) = \int_0^1 [f'(\xi(t)) - f'(a)] dt$$

Since $z, w \in D(a, r)$ we have

$$\left| \int_0^1 [f'(\xi(t)) - f'(a)] dt \right| \leq \int_0^1 |f'(\xi(t)) - f'(a)| dt < \varepsilon$$

In conclusion, if $z, w \in D(a, r)$, then $|g(z, w) - g(a, a)| < \varepsilon$. □

Theorem. (Rouche's theorem) If f and g are analytic inside and on a simple, closed contour C such that $|g(z)| < |f(z)|$ for every $z \in C$, then f and $f + g$ have the same number of zeros inside C .

Theorem. (Open mapping theorem) Let f be non-constant analytic in a domain Ω . If $D \subset \Omega$ is open, then $f(D)$ is open.

Proof. To show that $f(D)$ is open, take any $w_0 \in f(D)$. We need to find some $m > 0$ such that $D(w_0, m) \subset f(D)$. Let $f(z_0) = w_0$. Then there is $r > 0$ such that

$$\bar{D}(z_0, r) \subset D$$

and $f(z) - w_0$ has no zeros in $0 < |z - z_0| \leq r$ by the identity theorem. Let

$$m = \min_{|z - z_0| = r} |f(z) - w_0| > 0$$

Then we'll show that $D(w_0, m) \subset f(D)$. Take any $w_1 \in D(w_0, m)$ (we're trying to show that $w_1 \in f(D)$). Then

$$|w_0 - w_1| < |f(z) - w_0|$$

on $|z - z_0| = r$. Rouché's theorem implies that $f(z) - w_0$ and $f(z) - w_1$ have the same number of zeros inside $|z - z_0| = r$ which implies that

$$w_1 \in f(D(z_0, r)) \subset f(D)$$

□

Theorem. If f is one-to-one analytic in a domain D , then $f'(z) \neq 0$ for any $z \in D$.

Proof. Suppose that $f'(z_0) = 0$ for some $z_0 \in D$ (we'll show that f is not one-to-one). Then $f(z) - f(z_0) = f''(z_0)(z - z_0)^2 + \dots$ in some neighborhood of z_0 . There is $r > 0$ such that $\bar{D}(z_0, r) \subset D$ and $f(z) - f(z_0)$ and $f'(z)$ both have no zeros in $0 < |z - z_0| \leq r$.

Let $m = \min_{|z - z_0| = r} |f(z) - f(z_0)|$ and choose any $w_1 \in D(f(z_0), m)$. Then,

$$|f(z_0) - w_1| < m \leq |f(z) - f(z_0)|$$

on $|z - z_0| = r$. By Rouché's theorem, $f(z) - f(z_0)$ and $f(z) - w_1$ have the same number of zeros in $D(z_0, r)$ where $f(z) - f(z_0)$ has at least 2 zeros. In other words, since $f' \neq 0$ in $D(z_0, r)$, there is at least 2 distinct zeros of $f(z) - w_1$ in $D(z_0, r)$ where f is not one-to-one. □

LECTURE 8 (SEPTEMBER 30TH)

Theorem. Let $\{f_n\}$ be a sequence of 1-1 analytic functions on a domain D such that $f_n \rightarrow f$ uniformly on every compact subset of D . Then, either f is constant or f is 1-1 on D .

Proof. Let f be non-constant. Suppose that there exists $z_1, z_2 \in D$ such that $f(z_1) = f(z_2) = a$ and that the function is not 1-1. Then, there exists $r > 0$ such that:

(i) $\bar{D}(z_1, r) \subset D$ and $\bar{D}(z_2, r) \subset D$

(ii) By the identity theorem, $f(z) - a$ has no zeros in both $0 < |z - z_1| \leq r$ and $0 < |z - z_2| \leq r$. Denote simple circles $C_1 : |z - z_1| = r$ and $C_2 : |z - z_2| = r$. Let

$$m = \min_{z \in C_1 \cup C_2} |f(z) - a| \geq 0$$

where $C_1 \cup C_2$ is compact.

(iii) $\bar{D}(z_1, r) \cap \bar{D}(z_2, r) = \emptyset$

By the uniform convergence of $\{f_n\}$ on $C_1 \cup C_2$, there is N such that

$$|f_N(z) - f(z)| < m$$

for $z \in C_1 \cup C_2$. On C_1 , $|f_N(z) - f(z)| < m \leq |f(z) - a|$. By Rouché's theorem, $f(z) - a$ and $f_N(z) - a$ have the same numbers of zero inside C_1 . Also, $f(z) - a$ and $f_N(z) - a$ have the same number of zeros inside C_2 . Thus, $f_N(z) - a$ has zeros inside C_1 and inside C_2 . Thus, f_N is not 1-1. $\square$

Example. There are multiple cases where the first option holds. For example, take $f_n(z) = \frac{z}{n}$ and $f = 0$.

Theorem. (Riemann mapping theorem) If $D (\neq \mathbf{C})$ is a simply connected domain, then there is a 1-1 analytic function h from D onto U .

Proof. Let $\Sigma = \{f : D \rightarrow U \mid f \text{ is injective and analytic}\}$.

(I) We prove that $\Sigma \neq \emptyset$. To prove this, we use the prior result that if f is analytic on a simply connected domain D , such that $f(z) \neq 0$ for any $z \in D$, then there is g analytic in D such that $g^2 = f$ on D . Suppose that $w_0 \notin D$. Then there is g analytic on D such that $g^2(z) = z - w_0$. As $w^2 = z$ implies that $w = \sqrt{r}e^{i\theta/2}$, $g(D)$ does not intersect some half of the plane! Furthermore,

(i) $0 \notin g(D)$, which is trivial.

- (ii) g is 1-1 on D , as if $g(z_1) = g(z_2)$, then $g^2(z_1) = g^2(z_2)$ and $z_1 - w_0 = z_2 - w_0$.
- (iii) If $w \in g(D)$, then $-w \notin g(D)$, as if $g(z_1) = -g(z_2)$ for distinct points, then $g^2(z_1) = g(z_2)^2$ and $z_1 = z_2$ which is a contradiction.
- (iv) If $a \in g(D)$, then there exists an $r > 0$ such that $D(a, r) \subset g(D)$ due to the open mapping theorem. In particular, $D(-a, r) \cap g(D) = \emptyset$ as if $w \in D(-a, r)$, then $|w + a| < r$ and $|-w - a| = |w + a| < r$ which implies that $-w \in D(a, r) \subset g(D)$.

Now, define $h : D \rightarrow \mathbb{C}$ by

$$h(z) = \frac{r}{g(z) + a}$$

Notice that $|g(z) + a| > r$ for all $z \in D$ and that $|h(z)| < 1$ for all $z \in D$. We see that if $h(z_1) = h(z_2)$.

$$\frac{r}{g(z_1) + a} = \frac{r}{g(z_2) + a}$$

and that $g(z_1) = g(z_2)$ with $z_1 = z_2$. We see that $h(z)$ is a 1-1 function from an arbitrary domain D to U !

(II) Let $\psi \in \Sigma$ such that $\psi(D) \neq U$. We claim that for every $z_0 \in D$, there is $\psi_0 \in \Sigma$ such that $|\psi'_0(z_0)| > |\psi'(z_0)|$. Suppose that $\alpha \in U \setminus \psi(D)$. Then, $\phi_\alpha \circ \psi$ is analytic on D with no zeros. Since D is simply connected, there is g analytic in D such that $g^2 = \phi_\alpha \circ \psi$. Let $g(z_0) = \beta$. Define $\psi_0 = \phi_\beta \circ g$. Then, $\psi_0 \in \Sigma$ and due to the Schwartz lemma, the claim holds. $\square$

LECTURE 9 (OCTOBER 2ND)

Proof. (I) Take any $w_0 \notin D$, and then there is g that is analytic in D such that $g^2(z) = z - w_0$. If $D(a, r) \subset g(D)$, then $D(-a, r) \cap g(D) = \emptyset$. Now define

$$h(z) = \frac{r}{g(z) + a}$$

then $h \in \Sigma$.

(II) We claim that if $\psi \in \Sigma$ and $\psi(D) \neq U$, then for every $z_0 \in D$, there is $\psi_0 \in \Sigma$ such that $|\psi'_0(z_0)| > |\psi'(z_0)|$. Let $\alpha \notin \psi(D)$, then $\phi_\alpha \circ \psi$ is analytic in D with no zeros in D . Thus, there is g analytic in D such that $g^2 = \phi_\alpha \circ \psi$ in D . Given $z_0 \in D$, if $\beta = g(z_0)$, then $\psi_0 = \phi_\beta \circ g$ satisfies the claim.

Notice how $g \in \Sigma$ since g is 1-1 in D (if $g(z_1) = g(z_2)$, then $(\phi_\alpha \circ \psi)(z_1) = (\phi_\alpha \circ \psi)(z_2)$ concluding that $z_1 = z_2$). Additionally denote the square function by $s(z) = z^2$. Then,

$$\phi_\alpha \circ \psi = g^2 = s \circ g$$

so that $\psi = \phi_{-\alpha} \circ s \circ g$. Additionally, $\psi_0 = \phi_\beta \circ g$ so that $g = \phi_{-\beta} \circ \psi_0$. Therefore,

$$\psi = \phi_{-\alpha} \circ s \circ \phi_{-\beta} \circ \psi_0$$

on D . Now we will show that the claimed inequality holds. Defining $F = \phi_\alpha \circ s \circ \phi_{-\beta}$ or $F : U \rightarrow U$ is analytic. By the Schwartz lemma,

$$|F'(0)| \leq 1$$

since F is not 1-1. From $\psi = F \circ \psi_0$, we have

$$\psi'(z_0) = F'(\psi_0(z_0))\psi_0'(z_0) = F'(0)\psi_0'(z_0)$$

Since ψ is 1-1, the derivative $\psi' \neq 0$ in D and we have proved the claim.

(III) We now claim that if we take any $z_0 \in D$ and let $\eta_0 = \sup\{|\psi'(z_0)| \mid \psi \in \Sigma\}$, there is $h \in \Sigma$ such that $|h'(z_0)| = \eta_0$. From this, as $h(D)$ doesn't conform to the condition of step (II), we will have that $h(D) = U$. The key method here is using "normal families". $\square$

Remark. On $\mathbf{R}^n$,

- (i) If G is a bounded infinite, then G has a limit point
- (ii) If $\{a_n\}$ is a bounded sequence, then $\{a_n\}$ has a convergent subsequence
- (iii) If K is closed and bounded, then every open cover of K has a finite subcover

However, consider the following. The Hilbert space $l^2(\mathbf{N})$ can be seen as the space closest to $\mathbf{R}^\infty$.

$$\mathbf{x} = (x_1, x_2, \dots) \in l^2(\mathbf{N}) \quad \text{if} \quad \|\mathbf{x}\|^2 = \sum_{n=1}^{\infty} x_n^2 < \infty$$

Defined this manner, $l^2(\mathbf{N})$ is a complete normed vector space. Consider $E = \{\mathbf{e}_n \mid n \in \mathbf{N}\} \subset l^2(\mathbf{N})$ where $\mathbf{e}_k = (0, \dots, 0, 1, 0, \dots)$ where only the k -th element is nonzero. This basis satisfies $\|\mathbf{e}_k\| = 1$.

- (i) Then, E is a bounded infinite subset of $l^2(\mathbf{N})$ which has no limit point.
- (ii) Also, E is a bounded sequence in $l^2(\mathbf{N})$ with no convergent subsequence.
- (iii) Lastly, E is closed and bounded in $l^2(\mathbf{N})$ but is not compact.

Definition. (Three types of convergences & boundednesses) If D is open and $f_n : D \rightarrow \mathbf{C}$, then we can define three types of convergences.

- (i) Pointwise convergence in D
- (ii) Uniform convergence in D

(iii) Uniform convergence in every compact subset of D

There are also three types of boundednesses of $\{f_n\}$ on D

- (i) Uniformly bounded on D , meaning that $|f_n| \leq M$ for all $n \in \mathbf{N}$ and $z \in D$.
- (ii) Pointwise bounded on D , meaning there is $M_z > 0$ such that $|f_n(z)| \leq M_z$ for every $n \in \mathbf{N}$.
- (iii) Uniformly bounded on every compact subset of D , meaning that for every compact subset K of D , there is $M_K > 0$ such that $|f(z)| \leq M_K$ for all $n \in \mathbf{N}$ and $z \in K$.

Theorem. (Arzela-Ascoli theorem) A separable metric space is space that has a countable dense subset and a metric space is a space endowed with a metric. If $\{f_n\}$ is a pointwise bounded and equicontinuous sequence of functions on a separable metric space, $\{f_n\}$ has a subsequence that converges uniformly on all compact subsets.